Geometric and Learned Relay Control for Connectivity-Constrained Drone Delivery in Disaster Response

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Abstract

In disaster relief, delivery drones must stay connected to a depot through a multi-hop relay chain formed by the fleet itself, while carrying supplies with limited capacity and returning to reload. We model this as a makespan-minimisation problem in which every drone can either deliver or relay, and connectivity is a hard constraint enforced at every time step. We propose two controllers that share the same connectivity-preserving projection: a geometric chain-relay heuristic with a deadlock-recovery mechanism, and a fleet-size-agnostic policy learned once with discrete soft actor-critic over a set-based, autoregressive manager. Both are compared on identical instances with a paired protocol across fleets of 3 to 12 drones and 20 to 400 demand points, against four learning-free baselines and a time-expanded mixed-integer reference model on small instances. The learned policy completes every instance in every configuration; the heuristic completes 98–100% with the recovery mechanism and 62% without it. The two controllers divide the configuration space: with three drones the heuristic is 30–39 steps faster in paired median, with four drones the two are statistically indistinguishable over 480 paired runs, and with five or more drones the learned policy is faster by 38 to 178 steps. The policy is trained only on three and four drones and transfers without retraining to every larger fleet tested; the set-based architecture carries completion and training on randomised configurations carries speed. The three-drone deficit is localised to a single decision, which drone relays, and eleven policy-side interventions covering information, credit assignment, decision structure, initialisation and reward shaping all fail to remove it; half of it disappears when the connectivity projection is changed from radial shrinking to tangential sliding, showing that the deficit is partly a property of how the constraint is enforced rather than of the learner. On small instances solved to proven optimality, the learned policy is within 2–17% and the heuristic within 11–23% of the optimum of the space–time model on its finest lattice. Both controllers run in milliseconds per decision, three to six orders of magnitude below exact optimisation, which no longer proves optimality once reloading is needed.

Keywords: drone delivery, relay network, connectivity constraint, multi-agent reinforcement learning, heuristic, disaster logistics

1. Introduction

When communication, logistics and road infrastructure are destroyed, a fleet of drones can carry emergency supplies from a temporary depot to many demand points. Drones have been studied for last-mile distribution in disaster relief [1, 2, 3] and for medical delivery [4, 5], and the routing of drone fleets has grown into a literature of its own since the flying-sidekick travelling salesman problem [6, 7, 8, 9]. Two features make the disaster setting different from parcel delivery. First, every drone must be reachable by the depot at all times, because loading and unloading are supervised remotely and the delivery status must be tracked; with the ground network gone, reachability is provided only by the drones themselves acting as radio relays in a multi-hop chain. Second, the fleet is small and each drone carries a few units, so drones shuttle between the depot and the demand points and every drone that stands still as a relay is a drone that is not delivering. The operational question is how to assign delivery and relay roles over time so that the last demand point is served as early as possible without ever breaking the chain. Aerial relays are well studied in the communications literature [10, 11, 12], and connectivity maintenance is a classical topic in multi-robot control [13, 14], but in both the relay *is* the mission; here relaying competes with the delivery mission for the same vehicles.

Three properties make the problem hard. A hard connectivity constraint creates deadlocks: a greedy assignment that sends every drone to its nearest demand point completes none of the standard instances in our study, because the fleet pulls itself apart until no drone can move. Relaying competes directly with throughput, so the right number and identity of relays changes with the geometry of the remaining demand points and with how many drones are left with cargo. And exact optimisation does not scale: a space–time network model [15, 16] on a 98-metre lattice is solved to proven optimality in seconds while no reloading is needed, but once drones must reload the mixed-integer solver proves optimality on few instances within an hour and returns no feasible solution on some, whereas drone-routing models without a connectivity constraint are solved exactly for tens of customers [17].

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Proposal. We propose two controllers for this problem and study them side by side.

1. A *geometric chain-relay heuristic*. The drone farthest from the depot with cargo is the lead; the number of relays is the number of communication hops its target needs; the relays are the next-farthest drones, stationed on the segment from the depot to the lead at intervals of $0.9R$. Because the rule is deterministic it can repeat a deadlocked assignment indefinitely, so we add a *deadlock-recovery* mechanism that swaps the lead whenever the fleet has stalled for a fixed number of steps. The recovery mechanism alone lifts completion on the standard configuration from 62% to 98%.
2. A *learned policy* that is agnostic to fleet size. A manager assigns each drone one of seven discrete options every twenty steps; it encodes the drones, candidate demand points and relay slot as a set, pools them with attention, and decides autoregressively so that a formation of one lead and k relays can be composed deliberately. It is trained once with discrete soft actor-critic [18] on fleets of three and four drones and then run without retraining on fleets of up to twelve, in the spirit of size-agnostic learned solvers for routing [19, 20].

Both controllers act through the same connectivity-preserving projection: each commanded move is scaled by the largest factor that keeps the fleet connected. Connectivity is therefore guaranteed by construction for both, and the empirical question is whether a controller makes progress under the constraint rather than whether it violates it.

Research questions. We organise the study around five questions.

RQ1 Do the proposed controllers always complete the delivery while connectivity is enforced at every step, and when they do not, why not?

RQ2 On small instances, how close are the proposed controllers to the optimal solution of a space–time network model solved exactly?

RQ3 How does the performance of the two controllers compare across operating conditions, in particular across fleet size, number of demand points, depot position and communication radius, and where do the differences come from?

RQ4 Does the learned policy, trained on fleets of three and four drones, transfer upward to larger fleets and larger instances without retraining, and what carries the transfer?

RQ5 Are the components of each controller necessary?

Findings. The learned policy completes every instance in every configuration tested; the heuristic completes 98–100% with the recovery mechanism (RQ1). On instances small enough to solve to proven optimality, the learned policy is within 2–17% and the heuristic within 11–23% of the optimum of the space–time model on its finest lattice, and refining the lattice lowers that optimum by 7–9%, confirming that it bounds the continuous optimum from above (RQ2). The two controllers divide the configuration space by fleet size: the heuristic is faster with three drones, the two are statistically indistinguishable with four, and the learned policy is faster with five or more, by 62 to 178 steps up to ten drones and 76 at twelve; the three-drone deficit is a single decision, which drone relays, and it resists eleven policy-side interventions but shrinks by half when the connectivity projection is changed (RQ3). The learned policy never saw more than four drones or 80 demand points in training, yet it completes every instance with up to twelve drones and 400 demand points; the set-based architecture carries completion and the configuration randomisation in training carries speed (RQ4). Removing rule regularisation, configuration randomisation or entropy annealing from the learned policy costs hundreds of steps or completion, and removing the recovery mechanism from the heuristic costs a third of its completions (RQ5).

Contributions. (i) A geometric chain-relay heuristic with deadlock recovery and a fleet-size-agnostic learned policy for connectivity-constrained delivery, both built on a projection that makes the constraint hold by construction. (ii) A paired evaluation across fleet sizes that shows the two controllers are complementary, together with the mechanism behind the boundary, and evidence that the learned policy transfers upward to fleets three times larger than any it was trained on. (iii) A set of eleven documented negative results on closing the relay-limited deficit by changing the learner, and a sensitivity result showing that half of the deficit is attributable to the enforcement model. (iv) A space–time network model of the problem, solved exactly on small instances, with an explicit account of what its optimum does and does not bound.

2. Related Work

2.1. Vehicle routing with drones

The problem belongs to the family of vehicle routing problems [21, 22], with a homogeneous capacitated fleet, a single depot, unit demands and a makespan objective; the makespan (or min-max) objective, as opposed to total distance, is the standard choice when the goal is to finish service as early as possible [20, 23]. Drone routing began with the flying-sidekick travelling salesman problem, in which a truck and a drone cooperate [6], and was extended to multiple drones [24], to exact and approximate solution methods [25, 26, 17, 27], to vehicle routing with drones [28, 29, 30] and to drone-only fleets with energy and payload constraints [31]. The surveys of Otto et al. [7], Macrina et al. [8] and Chung et al. [9] map this literature. Across all of it communication is free: a drone is never required to be reachable from the depot, and no vehicle gives up its delivery role to serve as a relay. Our problem adds exactly that requirement, and the main consequence for modelling is that routes are no longer sufficient as decision variables, because the constraint concerns where all vehicles are at every instant (Section 3.1). Dynamic vehicle routing [32] shares the need to

decide online as the state evolves, which is the mode in which both of our controllers operate.

2.2. Drones in disaster and humanitarian logistics

Drones have been proposed for last-mile distribution when ground infrastructure is damaged. Chowdhury et al. [1] design a drone-based relief distribution network with a continuous approximation model; Rabta et al. [2] optimise a drone fleet for last-mile delivery under payload and range limits; Kim et al. [4] and Ghelichi et al. [5] plan drone fleets for medical deliveries in rural and urban settings; and Erdelj et al. [3] survey the roles drones can take in disaster management, including that of temporary communication nodes. These works plan at the strategic or tactical level (fleet size, depot location, route assignment) and assume that the control link exists. Our contribution is at the operational level: given a fleet and a depot, how the drones should move so that the link is maintained by the fleet itself while it delivers.

2.3. Aerial relays and connectivity maintenance

Using UAVs as communication relays is a mature topic in wireless networking: surveys cover flying ad-hoc networks [33], UAV communication issues [34] and UAV-enabled wireless networks [11]; throughput-optimal relay trajectories are derived in [10]; and relay placement for maintaining a link to a surveillance UAV is solved exactly in [12]. Multi-hop chains of mobile relays are studied in the robotics literature as communication chains [14], robotic routers [35] and relay positioning for team missions [36]. Connectivity maintenance for a team of mobile robots has a control-theoretic literature based on the algebraic connectivity of the communication graph [37, 13], and learned connectivity-preserving controllers exist [38]. Learning-based relay control has been applied to packet routing in UAV networks [39], to joint trajectory and power control of relay and mission UAVs [40] and to aerial relay networking [41]. In all of these the relays are the mission, or a dedicated sub-fleet with its own reward; they do not carry cargo, do not reload and do not minimise a delivery makespan. Switching between a task role and a relay role is an established idea in multi-robot exploration [42, 43], where the relay robots land and act as fixed nodes; our drones switch roles repeatedly and the relay positions move with the lead drone.

2.4. Time-expanded network models and symmetry

Our exact model is a time-indexed formulation on a space–time network. Time-expanded networks originate with Ford and Fulkerson's flows over time [15]; Skutella [16] surveys the theory. The dependence of such models on the time discretisation is well known, and dynamic discretisation discovery [44] refines the time grid where needed; we discretise space as well, which is the usual device in MILP trajectory planning for aircraft [45]. Connectivity is encoded as a single-commodity flow per period whose feasibility is equivalent to the existence of a multi-hop chain (Section 3.1). The drones are identical, so the model carries the usual index symmetry, which we break with the ordering constraints of Sherali and Smith [46]; Margot [47] surveys symmetry handling in integer programming.

2.5. Learning for routing and cooperative control

Neural solvers for routing problems, from pointer networks [48] to attention-based constructive policies [49, 19], show that set-based encoders with attention generalise across instance sizes; Bengio et al. [50] and Mazyavkina et al. [51] survey machine learning for combinatorial optimisation. Learned solvers for the min-max multiple travelling salesman problem [20, 52] are the closest in objective to ours, and we borrow the size-agnostic set encoding; none of them has a connectivity constraint, and none decides online. Dual-mission fleets in which the same drones deliver and sense [53] formalise a per-delivery detour budget, but the secondary task there is sensing rather than communication and it does not couple the vehicles' positions. On the learning side we build on soft actor-critic [18], centralised critics for decentralised execution [54, 55], value decomposition [56, 57], cooperative multi-agent policy optimisation [58] (see [59] for a review), temporal abstraction through options [60, 61, 62], invalid-action masking [63] and safe-action projection [64]. Our negative results in Section 6.3 test several of these ideas directly: per-agent credit, joint-action critics, monotonic value mixing and potential-style shaping [65] are all tried as remedies for the relay-limited deficit and none of them removes it.

2.6. Position of this work

To the best of our knowledge, no prior work combines (i) a homogeneous fleet carrying physical cargo to many demand points with capacity and reloading, (ii) a hard requirement that every drone be connected to the depot by a multi-hop chain at every step, and (iii) a makespan objective. In this setting relay duty competes with cargo throughput and the fleet must return to the depot repeatedly, which is what makes role assignment a sequential decision rather than a one-off placement. We contribute an exact space–time network model of this problem, a constructive online heuristic with a deadlock-recovery mechanism, and a learned size-agnostic policy, and we report where each is preferable.

3. Problem Statement

A depot sits at $c \in [0, 1000]^2$ metres and M demand points $d_1, \ldots, d_M$ lie within $0.9\,NR$ of it, where N is the number of drones and $R = 300$ m the communication range. Each drone flies at $v = 13.89$ m/s and carries $Q = 5$ units. Every delivery and every reload incurs a *service time*: unloading one unit at a demand point takes $S = 10$ steps and reloading to Q at the depot takes S steps, and during these steps the drone is stationary. A demand point or the depot is reached when a drone is within 25 m of it, and one step is one second. The service times matter for the connectivity constraint below: a stationary drone cannot move to keep a link alive, so its teammates must keep the chain intact around it. Every demand point needs exactly one unit. Drones do not collide, are not affected by wind, and battery is not modelled.

Connectivity. Let G_t be the graph whose vertices are the depot and the N drones and whose edges join vertices within R of each other. The fleet is *connected* at step t if G_t is connected, which is the same as saying that every drone has a multi-hop path to the depot. The constraint is that G_t is connected at every t. It is enforced rather than penalised (Section 4.1).

Objective. The makespan is the step at which the last demand point is served. A controller *completes* an instance if it serves all demand points within a budget of 10,000 steps. We report completion rate and, over completed instances, makespan.

Configurations. A configuration is a triple of fleet size N, number of demand points M and depot placement (fixed at the map centre, or uniformly random). The *standard* configuration is $N = 4$, $M = 50$ with the fixed depot. The learned policy is trained with $N \in \{3, 4\}$, $M \in [20, 80]$ and a random depot; all other configurations are unseen by it. The heuristic has no training.

3.1. A space–time network model, exact on the discretisation

The third element we propose, besides the two controllers, is an optimisation model of the problem. It serves two purposes: it states the problem without ambiguity, and on small instances it provides a plan obtained by exact optimisation against which both controllers are measured.

Why the model takes this form. Vehicle-routing formulations of drone delivery [6, 25] assign routes to vehicles and never need to know where a vehicle is at a given instant. The connectivity requirement of Section 3 is different in kind: it is a constraint on the *positions of all drones at every step*, because the communication graph G_t is built from those positions. A model must therefore carry every drone's position over time, and the natural object for that is a *time-expanded network* [15]: the set of positions is copied once per time period and arcs join a position in one period to the positions reachable in the next. Positions in the plane are uncountable, so space is discretised as well, to a lattice; the result is a *space–time network* on which drones are unit flows, and a time-indexed MILP on this network is exact for the discretised problem. Connectivity is then a property of which lattice nodes are occupied in each period, and we encode it with a single-commodity flow per period that is feasible exactly when the occupied nodes form a multi-hop chain to the depot. The model is *exact* in the sense that its optimum is the true optimum of the discretised problem; its relation to the continuous problem is stated at the end of this section.

Discretisation. A period is $\kappa = S = 10$ steps, so a service occupies exactly one period and a manager decision of the learned policy spans two; the per-period flight range is $\ell = v\kappa = 138.9\,\text{m}$. Space is a square lattice G of spacing $h = \ell/\sqrt{2} = 98.2\,\text{m}$ over the bounding box of the instance, from which nodes farther than $0.9NR$ from the depot are deleted because no chain can reach them; the depot and every demand point are added as exact nodes, so the interaction test is exact there. Figure 1 shows the resulting node set for a reference instance. With this spacing all eight king moves are feasible in one period; the move set and the communication set are

$$A(i) = \{j \in N_{\text{v}} : \|p_i - p_j\| \le \ell\} \ni i, \tag{1}$$

$$E = \{(i, j) \in N_{\text{v}}^2 : i \ne j,\ \|p_i - p_j\| \le R\}. \tag{2}$$

Axis-aligned moves cover h rather than ℓ per period, so the lattice slows a drone by up to 29% in those directions; this discretisation error always acts against the model. The lattice must also be fine enough to hold a relay chain: serving a demand point at distance d needs the intermediate drones placed within the chain's slack $NR - d$, which requires roughly $h \lesssim (NR - d_{\max})/(N - 1)$; a 196 m lattice makes the reference instances infeasible although they are feasible in the plane, and 98 m does not.

Node pruning by the convex hull. Most lattice nodes inside the reach limit are never useful. Let H be the convex hull of the depot and the demand points. For any configuration of the fleet, projecting every drone that lies outside H onto H does not increase any pairwise distance, because projection onto a convex set is non-expansive; hence it preserves every communication link, every one-period move and every service, and the projected plan has the same makespan. The continuous problem therefore always has an optimal plan inside H, and relay positions outside the hull can be discarded without loss. On the lattice we keep the nodes within $h/\sqrt{2}$ of H, the half-diagonal of a cell, so that every point of H still has a kept node within one lattice step; this margin guards against the case in which the projection of a lattice node is not itself a lattice node. The pruning removes 45% of the nodes of the 34 reference instances on average (from 2,881 to 1,594 in total) and, on the instances where we can check, leaves the lattice optimum unchanged (Section 6.2). Figure 1 shows the pruned and kept nodes.

Notation. Table 1 lists the sets, parameters and decision variables. Drones are identical, so a drone index only distinguishes units; the symmetry-breaking cut at the end of the model exploits this. The model is given in (3)–(16).

Reading the model. The objective (3) minimises the makespan C; the second term is a tie-break with weight $\varepsilon\bar{T}N \ll 1$ that prefers plans keeping idle drones at the depot and cannot change the optimal makespan.

Position and movement, (4)–(6). Every drone is at exactly one node at every period boundary and starts at the depot. A drone may be at node i at boundary $t + 1$ only if at boundary t it was at i itself or at a node from which i is reachable within one period. Because every move takes exactly one period, the position of every drone is defined at every boundary, which is what the connectivity constraints below need.

Deliveries, cargo and reloading, (7)–(12). Each demand point is served exactly once (7). A delivery during period t requires the drone to be at the demand point at both ends of the period (8), which is the service time of Section 3: the drone stands still for one period. Reloading is likewise a full period at

Table 1: Notation of the space–time network model.

Symbol	Domain	Meaning
Sets		
$N_v = \{0\} \cup D \cup G$		nodes: the depot (index 0), the demand-point nodes D, the lattice nodes G
K	$\lvert K\rvert = N$	drones
$T = \{0, 1, \ldots, \bar{T}\}$		period boundaries; period t runs from boundary t to $t + 1$
$A(i) \subseteq N_v$		nodes reachable from i within one period, including i (move set)
$E \subseteq N_v^2$		ordered pairs of distinct nodes within communication range R (communication arcs)
Parameters		
Q	5	cargo capacity (units)
R	300 m	communication range
κ, ℓ, h	10 steps, 138.9 m, 98.2 m	period length, flight range per period, lattice spacing
$\bar{T}$		horizon in periods, set to 1.15× the heuristic's makespan
ε	10^{-5}	tie-breaking weight
idx(j)		index of demand point j in a fixed ordering of D
Decision variables		
y_{ikt}	$\{0, 1\}$	1 if drone k is at node i at boundary t
s_{jkt}	$\{0, 1\}$	1 if drone k delivers to demand point j during period t
r_{kt}	$\{0, 1\}$	1 if drone k reloads at the depot during period t
q_{kt}	$\{0, \ldots, Q\}$	cargo on board drone k at boundary t
ρ_{kt}	$[0, Q]$	cargo taken on by drone k during period t
g_{ijt}	$[0, N]$	connectivity flow on arc (i, j) in period t
u_{it}	$\{0, \ldots, N\}$	occupancy of node i at boundary t, $u_{it} = \sum_k y_{ikt}$, with $u_{0t} := 1$
C	$[0, \bar{T}]$	makespan in periods

the depot (9). A drone does at most one service per period and can only deliver what it carries (10); cargo is tracked by (11) starting full; and a reload fills the drone completely (12), because ρ_{kt} can be positive only in a reload period and the next cargo must then be Q.

Multi-hop connectivity, (13)–(14). These two families are the part that distinguishes this model from drone-routing models. In every period, every occupied node ships one unit of flow, the depot is the only sink, and an arc may carry flow only if it leaves an occupied node. The system is feasible if and only if every occupied node is joined to the depot by a path whose intermediate nodes are occupied and whose consecutive distances are at most R, that is, if and only if the simulator's multi-hop connectivity test succeeds at that period boundary. The proof is short: if such paths exist, routing each node's unit along its path satisfies both families; conversely an unoccupied node $i \neq 0$ has $g_{ijt} = 0$ on all its outgoing arcs by (14) and therefore no incoming flow by (13), so all flow travels inside the occupied set and, since the depot is the only node without a conservation row, a flow decomposition yields the required paths. The mirror-image capacity rows $g_{ijt} \leq N u_{jt}$ are implied and omitted, which halves the dominant row count. Note what the constraint does to the problem: a drone that has nothing to deliver cannot simply wait at the centre if a teammate needs it as a hop, and a drone cannot fly to a demand point that would leave it farther than R from every occupied node. Relaying is therefore not a separate task but a consequence of where the fleet is.

Makespan and symmetry, (15)–(16). The makespan is at least the period of every delivery. Because drones are identical, any plan can be relabelled so that drone k serves no demand point with index below k; (16) imposes this ordering and removes no objective value.

What the optimum means. The optimum of the model is exact for the discretised problem. For the continuous problem of Section 3 it is an *upper bound*: drones are confined to lattice nodes and axis moves are shortened, so every lattice plan is a feasible continuous plan but not conversely, and a continuous controller can therefore be faster than the lattice optimum. The solver's dual bound is a bound on the lattice problem only. A certified lower bound through a cell relaxation, in which a lattice node stands for its cell and communication is tested by the minimum distance between cells, was attempted and abandoned: it inflates a three-hop chain's reach from 900 m to 1,317 m and returns bounds below the trivial analytic bound. We therefore call the solver's value a *reference plan*, replay every plan in the continuous simulator, and report the replayed makespan.

4. Proposed Controllers

Figure 2 summarises the setting and the two controllers. Both controllers decide at the level of *goals*: at every decision point each drone is given a goal (a demand point, the depot, or a relay slot) and flies towards it in a straight line at full speed until the next decision point. The two controllers differ only in how goals are chosen; the option set, the low-level flight, the relay-slot geometry and the connectivity projection are shared and are described first.

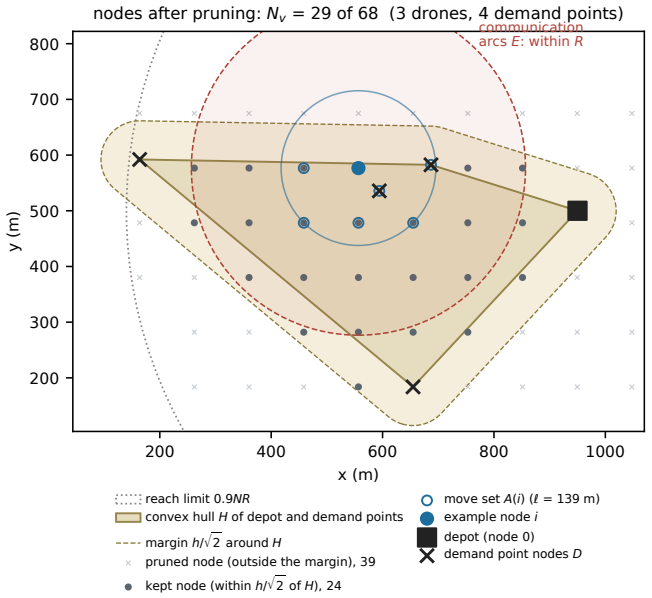


Figure 1: Nodes of the space–time network for a reference instance with three drones and four demand points: the depot (node 0), the demand-point nodes D, and the lattice G of spacing $h = 98.2$ m clipped to the reach limit $0.9NR$ (dotted). The shaded polygon is the convex hull H of the depot and the demand points and the lighter band is the margin of width $h/\sqrt{2}$ around it; lattice nodes inside the band or the hull are kept (dots) and those outside are pruned (crosses), leaving 29 of 68 nodes. For one example node i the figure marks its move set $A(i)$ (the nodes reachable in one period, $\ell = 138.9$ m) and its communication disc of radius R, which defines its arcs in E.

4.1. Shared interface: options, relay slots, projection

Decision points and options. A decision point occurs every 20 steps, and additionally as soon as any drone's goal becomes invalid: its demand point has been served by another drone, it has run out of cargo while heading to a demand point, it has become full while heading to the depot, or its relay slot has disappeared because the chain it belonged to shortened. At a decision point every drone receives one of $K = 7$ options: one of its five *candidate demand points*, which are the five unserved demand points nearest to it; *return* to the depot; or *relay*. An option is *valid* for a drone only if it makes sense at that moment: a demand point option requires cargo on board and an existing candidate, return requires the drone not to be full, and relay requires a free relay slot (defined next) and at least one teammate that is not itself relaying. Invalid options are masked for the learned policy and never produced by the heuristic; a drone with no valid option returns to the depot.

Relay slots. The environment, not the controller, decides where a relaying drone stands. For drone i the *frontier* f_i is the point farthest from the depot among the delivery goals of the other drones and the current positions of the other non-relaying drones (if there is none, the farthest unserved demand point). A chain to f_i at distance d needs $n = \min(\lceil d/(0.9R) \rceil - 1,\ N - 1)$ relay slots, placed on the segment from the depot to f_i at the fractions $1/(n+1), \ldots, n/(n+1)$ of its length; the factor 0.9 leaves slack so that a drone standing slightly off its slot does not break the link. Drone i takes the free slot nearest to it. A drone whose slot disappears holds position until the next decision point. The controllers therefore decide only *whether* a drone relays; the slot follows from the fleet's geometry.

Low-level flight and connectivity projection. Every step, each drone commands a displacement of length v towards its goal (zero while it is serving). Enforcing connectivity by a reward penalty would punish situations that no action can prevent, for example a drone that is stationary while unloading and is stranded when a teammate departs; connectivity is therefore enforced by projection. If the commanded positions keep G_t connected they are applied. Otherwise the projection repeatedly selects one moving drone, scales its displacement down to the largest factor in $[0, 1]$ under which the fleet is connected (found by bisection with the other drones' current scalings fixed), and stops as soon as the fleet is connected; at each round it selects the drone whose restriction leaves the most drones connected and breaks ties at random, so that no single anchor drone is starved. The previous positions are always connected, so zero scaling is a valid fallback and the constraint holds by construction. A drone whose displacement is scaled below 10% is *stalled* for that step, and a *partial deadlock* is a step in which all but at most one of the moving drones are stalled; the number of consecutive partial-deadlock steps, δ_t, is what the heuristic's recovery mechanism monitors. This projection scales a displacement along its own direction; Section 6.3 studies a variant that keeps the tangential component.

4.2. Geometric chain-relay heuristic with deadlock recovery

The heuristic is a deterministic rule evaluated at every decision point (Procedure 1). It first gives every drone a default: a drone with cargo goes to its nearest unserved demand point and a drone without cargo returns to the depot. It then selects the *lead*, the drone with cargo that is farthest from the depot, computes how many hops the lead's target needs, and turns that many of the remaining drones, the farthest ones first, into relays. The number of relays is adaptive and the relays are the drones already closest to where the chain must stand, which is what makes the rule cheap: it costs $O(N \log N)$ per decision and 0.05–0.12 ms in our implementation.

Deadlock recovery. Because the rule is deterministic, a geometry that stalls the fleet makes it re-issue the same assignment at every decision point and the stall persists; without an recovery mechanism the rule completes 62% of standard instances. The recovery mechanism is a timed override. It *starts* at the first decision point at which the fleet has been in partial deadlock for at least k_{stall} consecutive steps and no recovery is in progress. It *lasts* for h_{hold} decision points, that is, roughly $20\,h_{\text{hold}}$ steps, fewer when goals become invalid early; during these decision points the rule is evaluated as usual and then overridden as in step 6. It *ends* when the count e reaches zero, after which the

$$\min \; C + \varepsilon \sum_{i \in N_v \setminus \{0\}} \sum_{k \in K} \sum_{t \in T} y_{ikt} \tag{3}$$

$$\text{s.t.} \; \sum_{i \in N_v} y_{ikt} = 1 \qquad \forall k \in K,\ t \in T \tag{4}$$

$$y_{0k0} = 1 \qquad \forall k \in K \tag{5}$$

$$y_{i,k,t+1} \le \sum_{j \in A(i)} y_{jkt} \qquad \forall i \in N_v,\ k \in K,\ t < \bar{T} \tag{6}$$

$$\sum_{k \in K} \sum_{t < \bar{T}} s_{jkt} = 1 \qquad \forall j \in D \tag{7}$$

$$s_{jkt} \le y_{jkt}, \qquad s_{jkt} \le y_{j,k,t+1} \qquad \forall j \in D,\ k \in K,\ t < \bar{T} \tag{8}$$

$$r_{kt} \le y_{0kt}, \qquad r_{kt} \le y_{0,k,t+1} \qquad \forall k \in K,\ t < \bar{T} \tag{9}$$

$$\sum_{j \in D} s_{jkt} + r_{kt} \le 1, \qquad \sum_{j \in D} s_{jkt} \le q_{kt} \qquad \forall k \in K,\ t < \bar{T} \tag{10}$$

$$q_{k,t+1} = q_{kt} - \sum_{j \in D} s_{jkt} + \rho_{kt}, \qquad q_{k0} = Q \qquad \forall k \in K,\ t < \bar{T} \tag{11}$$

$$\rho_{kt} \le Q\, r_{kt}, \qquad q_{k,t+1} \ge Q\, r_{kt} \qquad \forall k \in K,\ t < \bar{T} \tag{12}$$

$$\sum_{j:(i,j) \in E} g_{ijt} \; - \sum_{j:(j,i) \in E} g_{jit} = u_{it} \qquad \forall i \in N_v \setminus \{0\},\ t \in T \tag{13}$$

$$g_{ijt} \le N\, u_{it} \qquad \forall (i,j) \in E,\ t \in T \tag{14}$$

$$C \ge t\, s_{jkt} \qquad \forall j \in D,\ k \in K,\ t < \bar{T} \tag{15}$$

$$s_{jkt} = 0 \qquad \forall k \in K,\ j \in D \text{ with } \mathrm{idx}(j) < k,\ t < \bar{T} \tag{16}$$

plain rule resumes; if the deadlock persists it can start again at the next decision point. Two modes are implemented. In *lead swap* the current lead is sent back to the depot and the role of lead passes to another drone; on the m-th recovery of an episode it passes to the m-th farthest drone, so that repeated recoveries cycle through the fleet rather than swapping the same two drones. In *retreat* every drone except the lead returns towards the depot, which slackens the chain. Section 6.5 selects the mode and the two parameters on tuning seeds; lead swap with $k_{\text{stall}} = 120$ and $h_{\text{hold}} = 20$ is used in every comparison.

4.3. Hierarchical learned policy

The learned controller is hierarchical (Figure 3). The high level, a *manager*, acts only at decision points and outputs one option per drone; the low level is the shared straight-line flight of Section 4.1, so the manager's choices are held for 20 steps and the problem it solves is a semi-Markov decision process over options [60]. Only the manager is learned. Separating the levels has two consequences that the rest of the design relies on: the manager sees the fleet at a coarse time scale at which a formation can be held, and the low level, being a fixed rule, can be reasoned about and projected for connectivity without learning.

Set-based encoder. The manager encodes drone i from four kinds of entities, all distances divided by R. *Own state*, 10 features: cargo fraction; distance to the depot; hop depth divided by 8; fraction of demand points unserved; elapsed time divided by the budget; its realised-movement ratio (an exponential average of achieved over commanded displacement); whether it could advance one full step towards its goal alone without disconnecting the fleet; the fleet-wide realised-movement ratio; service timer; and whether it is a cut vertex of G_t. *Candidate demand points*, five entities of 6 features: distance; the two components of the unit direction vector; whether a teammate already targets it; its distance from the depot; and the hops its chain requires, divided by 8. *Peers*, $N - 1$ entities of 9 features: distance; the two components of the relative position vector; cargo fraction; hop depth divided by 8; and a four-way one-hot of the peer's current goal kind (deliver, return, hold, relay). *Relay slot*, 4 features: distance; the two components of the unit direction vector; and the number of slots in the current chain, divided by 8. Each entity type passes through a two-layer MLP of width 128. The drone's own embedding is the query of two multi-head attention layers (4 heads) that pool the peer set and the candidate set into one vector each; the four vectors are concatenated and fused by a further MLP into the drone's embedding h_i. A *pointer head* scores each candidate from $[h_i; e_{\text{cand}}]$ and a *fixed head* scores return and relay from h_i; the seven scores are masked and normalised by a softmax. Nothing in the network depends on N or M, which is why one set of weights runs on fleets it never saw.

Autoregressive decision. The N options are not sampled independently. Drones are ordered by distance from the depot, farthest first; the manager samples the first drone's option, writes its new role into the role one-hot of the peer features that the remaining drones see, re-encodes, samples the next drone, and so on (Procedure 2). A formation of one lead and k relays arises only by coincidence under independent sampling but can be composed deliberately under sequential decisions; the cost is

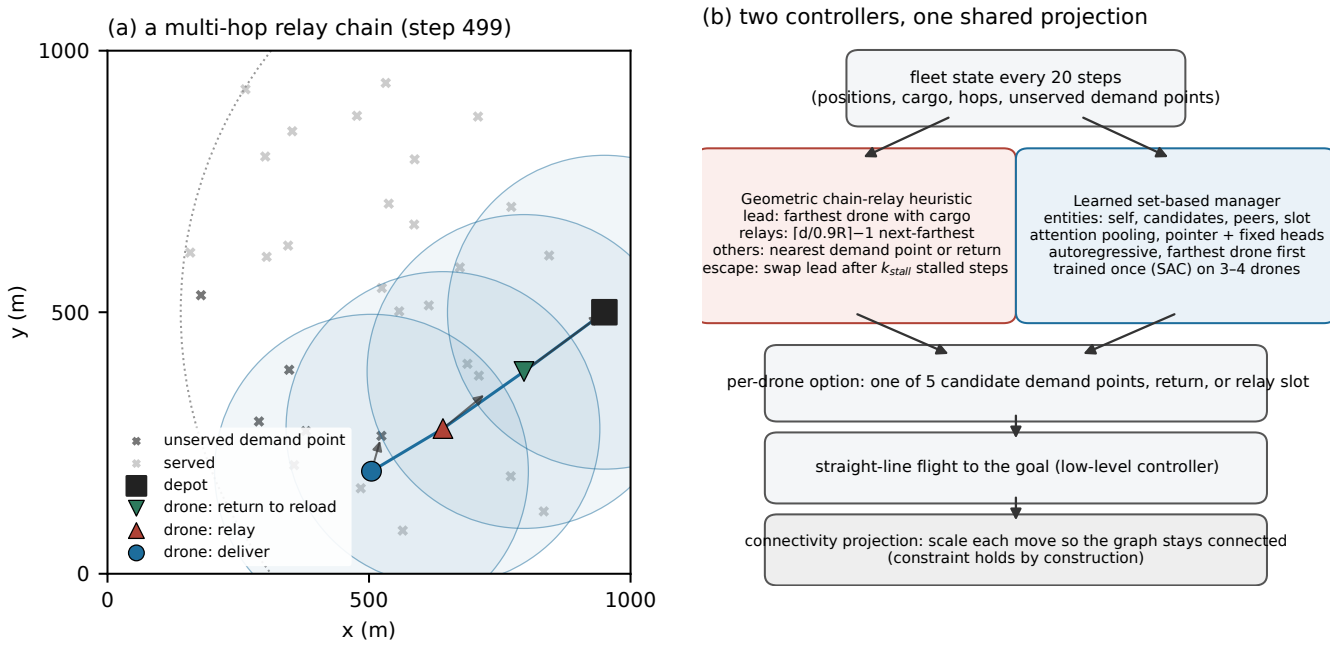


Figure 2: (a) An instance with three drones under the heuristic at step 499: the lead (blue circle) heads for a far demand point, one drone relays on the segment to the depot (red triangle), and one drone returns to the depot to reload (green triangle); shaded discs are communication ranges and the dotted circle is the reachable region $0.9NR$. (b) The two controllers emit the same per-drone options and share the straight-line low-level controller and the connectivity projection.

Procedure 1. Geometric chain-relay heuristic with deadlock recovery, run at every decision point. State kept between calls: remaining recovery length e (initially 0) and recovery count m (initially 0); both reset at the start of an episode. Parameters: $k_{\text{stall}} = 120$, $h_{\text{hold}} = 20$, mode $\in$ {lead swap, retreat}.

1. **Recovery trigger.** If the fleet has been in partial deadlock for $\delta_t \geq k_{\text{stall}}$ consecutive steps and $e = 0$, set $e \leftarrow h_{\text{hold}}$ and $m \leftarrow m + 1$.
2. **Default assignment.** For each drone i: if it has no cargo, option $\leftarrow$ return; else if it has no unserved candidate, option $\leftarrow$ relay; else option $\leftarrow$ its nearest unserved demand point.
3. **Lead.** If no demand point is unserved, stop. Otherwise let $r_i = \|p_i - c\|$ and let the lead be the drone with cargo that maximises r_i (if no drone has cargo, the drone maximising r_i).
4. **Relay count.** Let $d = \max(r_{\text{lead}}, \|d_{\text{goal}} - c\|)$ where d_{goal} is the lead's nearest unserved demand point, and $n = \min(\max(1, \lceil d/(0.9R) \rceil) - 1,\ N - 1)$.
5. **Relay identity.** Sort the drones other than the lead by r_i in decreasing order and set the option of the first n of them to relay.
6. **Recovery manoeuvre** (only if $e > 0$; then $e \leftarrow e - 1$). *Lead swap*: sort all drones by r_i in decreasing order, let the new lead be the $\min(m, N-1)$-th drone of that order (0-based), set the old lead's option to return and, if the new lead has cargo and an unserved candidate, set its option to its nearest unserved demand point. *Retreat*: set the option of every drone except the lead to return.
7. Return the options; the environment converts them into goals and assigns relay slots.

N sequential network calls per decision point.

Training. The manager is trained with discrete soft actor-critic [18] on option transitions. A transition stores the fleet observation at a decision point, the N options, the option return $\sum_\tau \gamma^\tau r_\tau$ accumulated over the steps until the next decision point, the observation at that point, and the discount $\gamma^{\Delta t}$ with $\gamma = 0.997$ per step. The team reward is $+10$ per delivery, $+30$ per reload, $-0.01N$ per step and a residual penalty of $-0.1N$ times the fraction of demand points still unserved. A twin-Q critic with attention across the drone embeddings outputs per-drone action values whose mean over drones is regressed on the option return; the actor minimises the standard expectation form $\mathbb{E}_{a\sim\pi}[\alpha \log \pi(a) - Q(a)]$ per drone, with teacher forcing of the autoregressive order. Three elements of the training configuration matter and are ablated in Section 6.5. (i) The entropy target is annealed linearly from $0.6 \log n_{\text{valid}}$ to $0.25 \log n_{\text{valid}}$ over the first 400 episodes: a high target keeps the policy random enough to recovery deadlocks early in training, a low one lets it hold a multi-step formation later. (ii) A cross-entropy term with weight 2.0 pulls the policy towards the option the heuristic's assignment rule (Procedure 1, steps 2–5, without recovery) would take, wherever that option is valid. (iii) Every training episode samples $N \in \{3, 4\}$, $M \in [20, 80]$ and a random depot. Optimisation uses Adam at 3×10^{-4}, batches of 256 transitions from a replay buffer of 60,000, one gradient step every 4 environment steps after a warm-up of 400 transitions, and Polyak averaging of the target critic with $\tau = 0.005$. Checkpoints are scored every 50 episodes on 80 validation instances with a 3,000-step budget by $100f + 40(1 - \bar{t}/3000)$, where f is the completion rate and $\bar{t}$ the mean steps; training stops after eight evaluations without improvement or at 1,000 episodes. One run takes about 13 hours on a single RTX 5080 when four runs share the GPU.

5. Experimental Setup

Instances and configurations. Instances are generated from integer seeds; a seed, N, M and the depot mode determine the instance exactly, so two controllers can be run on identical instances. Evaluation seeds are disjoint from the seeds used for training, for checkpoint selection and

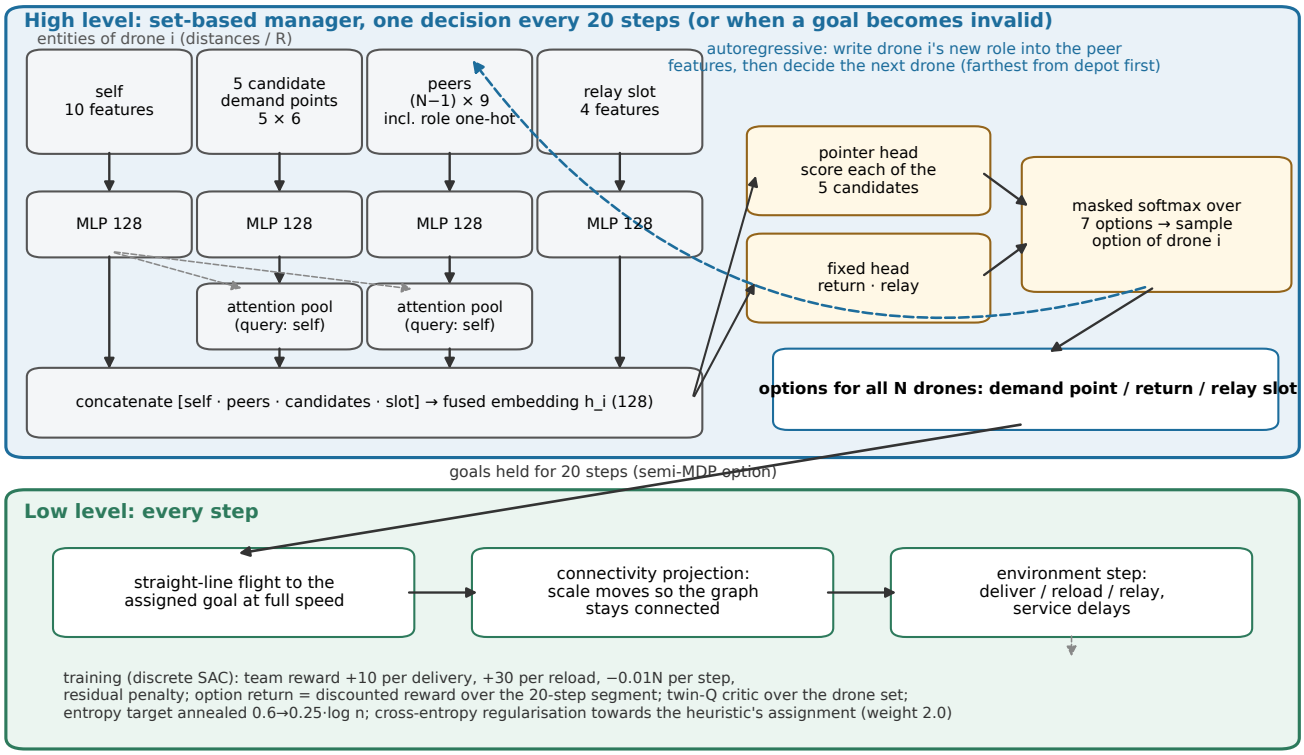


Figure 3: The hierarchical learned controller. High level: the set-based manager encodes each drone from four kinds of entities, pools the peer and candidate sets with attention queried by the drone's own embedding, and scores the five candidates with a pointer head and return/relay with a fixed head; drones are decided one at a time, farthest from the depot first, and each decision is written into the peer features seen by the next drone. Low level, every step: straight-line flight to the assigned goal, the connectivity projection, and the environment step. Options are held for 20 steps.

Procedure 2. Learned manager at a decision point (inference).

1. Build the entity observation of every drone from the fleet state; compute the validity mask of the seven options for every drone (Section 4.1).
2. Order the drones by distance from the depot, farthest first.
3. For each drone i in that order: encode all entities with the current peer roles, compute the seven masked logits of drone i, sample its option from the softmax (sampling, not argmax, is the deployed inference), and write the chosen role (deliver, return or relay) into the peer role one-hot of drone i as seen by the drones not yet decided.
4. Return the N options; the environment converts them into goals and assigns relay slots.

for tuning the heuristic. We evaluate on the standard configuration ($N = 4$, $M = 50$, fixed depot) and on eight unseen configurations with random depots: $N/M \in \{3/20, 3/30, 4/80, 5/50, 6/80, 8/150, 10/300, 12/400\}$.

Baselines. Four learning-free baselines establish that the problem is not solved by naive assignment. All use the shared interface of Section 4.1 and are evaluated at the same decision points. *Greedy, no relay* is step 2 of Procedure 1 alone: every drone with cargo flies to its nearest unserved demand point and every empty drone returns; nobody ever relays, so the chain is maintained only by the projection, which stalls the fleet. *One (two) fixed relays* assigns the first one (two) drones of the fleet to relay permanently and runs the greedy rule on the others; the relay count is fixed in advance instead of adapting to the lead's distance. *Geometric rule, no recovery* is Procedure 1 without step 6. All learning-free controllers are deterministic except for the random tie-breaking of the projection.

Reference model. Instances with $M \leq 5$ demand points are solved to proven optimality with the breadth-first search of Section 6.2 on the hull-pruned lattice; instances with $M \geq 6$ are solved with the space–time MILP in Gurobi 13 [66], four threads and one hour per instance. Every optimal or best-found plan is replayed in the continuous simulator.

Paired protocol. Averaging only the episodes each method completes favours the method that gives up on hard instances, and a single long stall moves a 40-run mean by over 100 steps. All comparisons between the two controllers are therefore *paired*: both run on the same instance, the difference is taken over instances both complete, and we report the median paired difference, the win–loss count and a Wilcoxon signed-rank p-value. Inference samples from the policy; deterministic argmax inference has the same median but a far heavier tail. The standard configuration uses 480 runs (60 seeds, 8 repetitions, two independent measurement runs), the 3/30 and 5/50 configurations 200 (40 seeds, 5 repetitions) and the remaining configurations 80 (40 seeds, 2 repetitions). Repetitions differ because the projection's tie breaking is random. We found 40 runs too few to support any claim about tails.

Table 2: Completion within the 10,000-step budget. Learning-free baselines on 120 (standard) or 40 (unseen) runs; the two proposed controllers on the paired run counts of Section 5. p is Fisher's exact test (two-sided) of each method's completion count against the learned policy on the same configuration.

Method	4/50 std.	p	3/30	p	5/50	p
Greedy, no relay	0%	$< 10^{-4}$	0%	$< 10^{-4}$	25%	$< 10^{-4}$
One fixed relay	0%	$< 10^{-4}$	2.5%	$< 10^{-4}$	40%	$< 10^{-4}$
Two fixed relays	3.3%	$< 10^{-4}$	52.5%	$< 10^{-4}$	65%	$< 10^{-4}$
Rule, no recovery	61.7%	$< 10^{-4}$	95%	0.027	92.5%	0.004
Rule + recov. (prop.)	473/480	0.015	200/200	1	191/200	0.004
Learned (prop.)	**480/480**	—	**200/200**	—	**200/200**	—
	3/20		4/80		6/80	
Rule + recov. (prop.)	80/80	1	80/80	1	79/80	1
Learned (prop.)	**80/80**		**80/80**		**80/80**	
	8/150		10/300		12/400	
Rule + recov. (prop.)	80/80	1	80/80	1	80/80	1
Learned (prop.)	**80/80**		**80/80**		**80/80**	

6. Results

6.1. RQ1: Completion under the hard constraint

Connectivity was never violated in any training episode or evaluation run, as the projection guarantees. Table 2 reports completion within the 10,000-step budget. The learned policy completes every instance in every configuration. The heuristic with recovery completes 98–100%; its misses concentrate where drones are plentiful, where a long chain that is not needed still gets stationed and stalls. Without the recovery mechanism the rule completes 61.7% of standard instances, and the naive baselines complete almost none: the greedy assignment pulls the fleet apart, and fixed relay counts are wrong for most instances because the number of relays a demand point needs varies with its distance.

6.2. RQ2: Distance to the exact optimum of the space–time model

Exact solutions. For instances in which no reload is needed ($M \le Q = 5$) we solve the space–time model of Section 3.1 to proven optimality by breadth-first search over its state space: a state is the multiset of the three drones' nodes together with the set of served demand points, the fleet is identical so states are kept up to permutation, and the search expands all node triples reachable in one period that satisfy the multi-hop connectivity test, period by period, until every demand point is served. Because every transition is exactly a feasible column of the MILP, the first period at which the full set appears is the optimal makespan of the model. On the hull-pruned lattice this takes 0–10 seconds per instance at $h = 98$ m, which we verified against Gurobi on the instance it could certify (seed 902, ten periods, 1,242 s with the solver, 1 s with the search), and the pruned and unpruned lattices gave the same optimum on every instance we checked. The search also lets us refine the lattice: Table 3 reports the optimum at spacings $h = 98$, 69 and 49 m on ten instances each for $M = 3, 4, 5$; at 49 m one instance takes up to 52 minutes.

Result. Three observations follow (Table 3, Figure 4). First, the lattice optimum is an upper bound on the continuous optimum that decreases as the lattice is refined: the mean optimum falls by 7–9% from $h = 98$ to $h = 49$ m (89 to 82, 110 to 103 and 121 to 110 steps for $M = 3, 4, 5$). It is not monotone on every instance, because lattices of different spacing are not nested, but the finest lattice is the tightest reference we have and all ratios below are taken against it. Second, at $h = 49$ m the learned policy's median is 1.7%, 4.7% and 16.7% above the lattice optimum for $M = 3, 4, 5$, and the heuristic 12.6%, 10.7% and 22.7%; the learned policy is faster than the heuristic on the same instances for $M = 3$ and 4 (Wilcoxon $p = 0.016$ and 0.008 over ten instances) but not significantly for $M = 5$ ($p = 0.36$), where both controllers lose ground because five demand points with three drones force a second relay chain. The recovery override of Section 6.3 changes the learned policy's median by at most two percentage points here. Third, a plan that is optimal on the lattice is not optimal in the plane: replaying each optimal plan in the continuous simulator gives a makespan 2–10 steps below the lattice value on most instances, because the simulator moves continuously between the plan's nodes, and one plan ($M = 5$, instance 4) fails to replay because the continuous projection and the lattice's one-period moves do not coincide exactly. The remaining distance between the controllers and the optimum therefore has three parts: the period quantisation of ten steps, the lattice restriction that persists at 49 m, and the controllers' own suboptimality; the first two can be reduced only at a cost that grows with M (the search time rises from seconds at 98 m to 25 minutes on average at 49 m for $M = 5$).

Instances that need reloading. For $M \ge 6$ a drone must return to reload, the state space of the search grows by a cargo dimension per drone, and we fall back to the MILP with the hull-pruned lattice at $h = 98$ m, one hour per instance and four solver threads (Table 4). Here the solver's reach ends. With six demand points it returns a feasible solution on 9 of 10 instances and proves optimality on 3; with eight it returns a solution on 5 of 10 and proves none; with ten it returns one solution out of four. The mean remaining optimality gap grows from 14% to 31% and 37%, so the solver's incumbent is an upper bound on the lattice optimum of unknown quality rather than a reference. Measured against that incumbent the heuristic is 1.16, 1.11 and 1.08 times its makespan and the learned policy 1.07, 0.97 and 1.05; the learned policy beats the incumbent on the eight-demand-point instances, which only says that an hour of branch and bound no longer produces a good plan at that size. The comparison in RQ3 is therefore between two controllers that, where an exact optimum is available, are within 2–17% (learned) and 11–23% (heuristic) of it, and the exact reference stops being available exactly where the controllers are needed.

6.3. RQ3: Comparative performance across operating conditions

Fleet size divides the configuration space. Table 5 and Figure 5 give the paired comparison across the nine configurations. With three drones the heuristic is faster by 39 and 30 steps in paired median with 20 and 30 demand points. With four drones the two controllers are statistically indistinguishable: on the standard configuration the median paired difference over 480 runs

Table 3: Exact optimum of the space–time model versus the proposed controllers on instances with three drones and $M \leq 5$ demand points, ten instances per M. Columns h give the mean lattice optimum (steps) at three lattice spacings; the ratio columns divide each controller's makespan on the continuous simulator (heuristic: deterministic; learned: median of 30 runs; learned + recovery: median of 30 runs with the deadlock-recovery override) by the optimum at $h = 49$ m, averaged over instances; p is a Wilcoxon signed-rank test of heuristic against learned median across the ten instances.

M	n	$h = 98$	$h = 69$	$h = 49$	heuristic	learned (median)	learned + recovery	p (heur. vs. learned)
3	10	89	87	82	1.126	1.017	1.035	0.016
4	10	110	107	103	1.107	1.047	1.048	0.008
5	10	121	118	110	1.227	1.167	1.161	0.355

Table 4: Instances that need reloading ($M \geq 6$): space–time MILP with the hull-pruned lattice ($h = 98$ m), one hour per instance. "feasible" counts instances with a feasible solution, "optimal" those with a proven optimum, "mean gap" the mean optimality gap over feasible solutions and "mean time" the mean solver time. The ratio columns divide the controllers' makespans by the best solution found, averaged over feasible instances.

M	instances	feasible	optimal	mean gap	mean time (min)	heuristic / best	learned / best	learned + recovery / best
6	10	9	3	14%	53	1.16	1.07	1.09
8	10	5	0	31%	60	1.11	0.97	0.98
10	4	1	0	37%	60	1.08	1.05	1.06

is +2 steps with 234 wins against 239 ($p = 0.52$), and on 4/80 the learned policy is 38 steps faster. With five or more drones the learned policy is faster, by 62 steps at five drones growing to 178 at ten and 76 at twelve (70 wins against 10). The boundary therefore lies at four drones, and the advantage persists beyond the largest fleet we measured, although the monotone growth stops between ten and twelve (Section 6.4 returns to this point). An earlier measurement against the heuristic's pre-tuning recovery parameters gave the same picture (+40, +0, −54, −170 on 3/30, 4/50, 5/50 and 10/300), so the boundary does not depend on the tuning.

Figure 6 shows the full distributions on the three main configurations: with three drones the heuristic's curve lies to the left, with four the curves coincide, and with five the learned policy's curve lies to the left and reaches completion earlier.

Tails follow the same split. Where drones are scarce the learned policy's worst case is three to four times the heuristic's (maximum 5,395 versus 1,447 steps on 3/30); where they are plentiful it is lighter (1,101 versus 1,734 on 5/50, 1,365 versus 1,753 on 10/300). On the standard configuration over 480 runs against the pre-tuning heuristic the learned policy exceeded 1,500 steps five times and the heuristic fourteen times.

Mechanism. The two controllers differ in one decision. On 3/30 the heuristic stations 1.39 relays on average and the learned policy 0.78; the learned fleet therefore spends 27% of drone-steps stalled by the projection against 8% for the heuristic. Overriding only the learned policy's relay assignment with the heuristic's, at inference, recovers the full 44-step gap; matching only the *number* of relays recovers 14 steps of median but removes the tail (maximum from 4,904 to 1,083). The identity of the relaying drone drives the median and the count drives the tail. The information needed to make the heuristic's choice is present in the learned policy's observation: a two-layer classifier predicts the heuristic's relay assignment from the 41 observed features with 85% accuracy, and behaviour cloning of the full rule reaches 95% on relay decisions. When drones are plentiful the opposite happens: the heuristic keeps stationing a chain that the geometry no longer needs, while the learned policy exploits the slack and, with its sampled inference, recoveries the stalls that the deterministic rule repeats; this is also where the heuristic loses its few completions.

Why the relay-limited deficit resists policy-side fixes. Table 6 lists eleven interventions, each trained with four or two seeds under the same training configuration except for the stated change and evaluated with the same paired protocol, that attempt to make the learned policy relay like the heuristic when drones are scarce. The variants are, in the numbering of the table: (V1) the cross-entropy regularisation towards the heuristic is weighted three times higher for drones whose heuristic option is relay; (V2) each drone on the communication path of a delivering teammate receives an individual reward (2.0 or 6.0) at that delivery and the critic is trained on per-drone returns; (V3) checkpoint selection scores half of its validation instances on the 3/30 configuration instead of only on 4/50, eight seeds; (V4) as V2 but zero-sum, the delivering drone losing what its relays gain; (V5) the critic is conditioned on the teammates' chosen roles (a joint-action critic) with the teammates' next roles sampled from the actor in the target; (V6) the autoregressive order is reversed to decide the drone nearest the depot first; (V7) all goals are invalidated after 60 consecutive partial-deadlock steps so that the manager re-decides, in training and evaluation; (V8) the mean over per-drone action values in the critic is replaced by state-dependent monotone weights that sum to one with a floor of one half; (V9) each drone's option is stored as its own single-agent transition, chained inside a decision point with discount one, so that no decomposition of the team return is needed; (V10) the manager is initialised by behaviour cloning of the heuristic on 40,000 decisions (92.5% accuracy, 95% on relay decisions) and then fine-tuned, with and without a 50-episode critic warm-up; (V11) a per-step penalty proportional to the estimated arrival time of the bottleneck demand point, with the estimate divided by each drone's realised-movement ratio so that blocked drones count

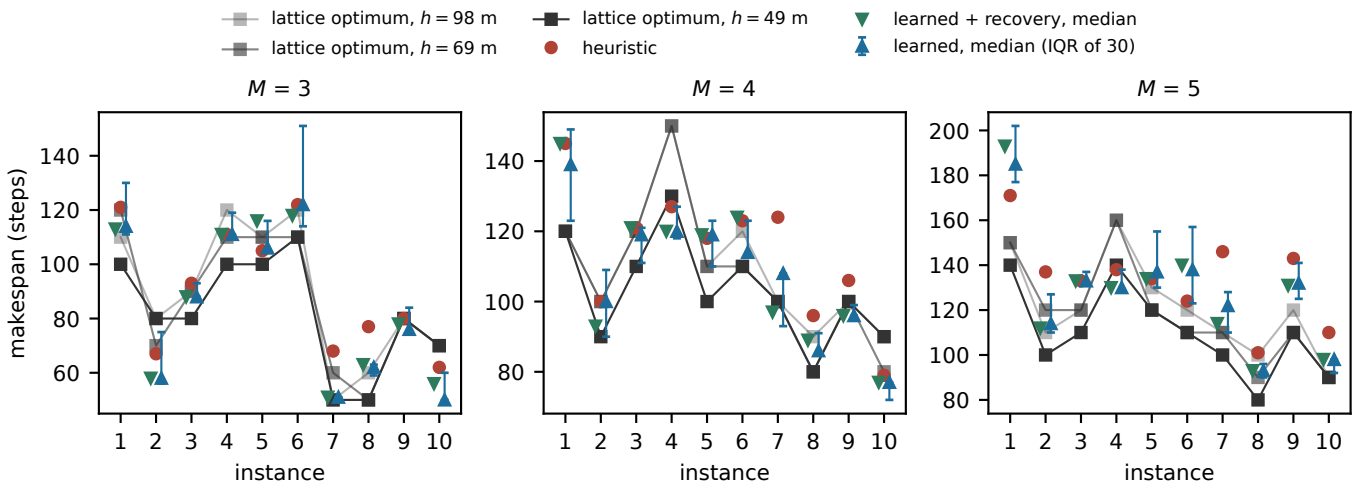


Figure 4: Per-instance makespan on the exactly solved instances with three drones: lattice optimum at three spacings (squares, darker is finer), heuristic (circle), learned policy median with the interquartile range of 30 runs (upward triangle) and learned policy with the recovery override (downward triangle).

Table 5: Paired comparison on identical instances. Positive differences mean the learned policy is slower than the heuristic. p from the Wilcoxon signed-rank test on both-completed pairs.

N/M	rollouts	complete h/l	median heuristic / learned	paired median	win–loss	p
3/20	80	80/80	364 / 420	+39	22–58	$< 10^{-4}$
3/30	200	200/200	572 / 618	+30	81–119	3.4e-04
4/50 (standard)	480	473/480	878 / 878	+2	234–239	0.521
4/80	80	80/80	1,120 / 1,036	−38	50–28	0.006
5/50	200	191/200	588 / 534	−62	147–44	$< 10^{-4}$
6/80	80	79/80	737 / 614	−122	72–7	$< 10^{-4}$
8/150	80	80/80	916 / 722	−158	75–5	$< 10^{-4}$
10/300	80	80/80	1,216 / 1,052	−178	69–11	$< 10^{-4}$
12/400	80	80/80	1,199 / 1,108	−76	70–10	$< 10^{-4}$

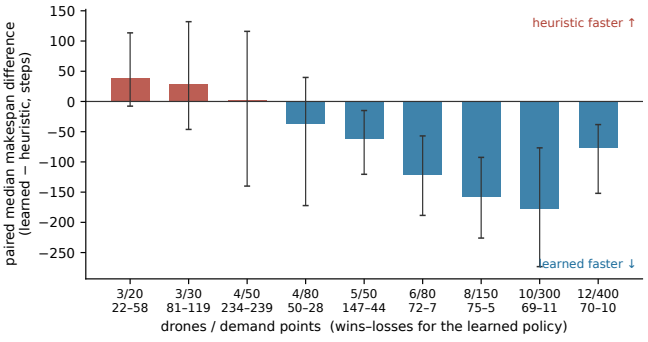


Figure 5: Paired median makespan difference (learned minus heuristic) by configuration, with the interquartile range of the paired differences as error bars and the learned policy's wins–losses under each label. Negative values favour the learned policy; the boundary lies at four drones.

as slow. None improves the 3/30 median and several hurt the configurations where the learned policy wins. The pattern is informative. Adding reward for relaying turns the means into the end: the policy relays for its own sake and 5/50 reverses from −54 to +60. Every route to per-drone credit is blocked for a different reason: a joint-action critic inherits the variance of a sampled teammate action in its target; a monotonic mixing network either diverges when its weights are free or collapses to the mean when they are floored; a sequential single-agent reformulation removes the decomposition but leaves an undiscounted credit chain with no contraction. Initialising the policy by behaviour cloning of the rule, with 95% relay accuracy, is erased within 400 episodes: once the actor is updated, relay usage falls to the level of a randomly initialised policy within 50 episodes and stays there. A penalty on the bottleneck demand point's estimated arrival makes the policy reduce the *average* rather than the maximum. We conclude that, within this policy structure and objective, under-relaying is the learner's fixed point, not an artefact of credit, information or initialisation.

Sensitivity to the connectivity enforcement model. The projection of Section 4.1 scales a blocked displacement along its own direction, so a drone pulling on the chain effectively stops. We implemented an alternative that clamps the target position to the nearest point of the communication disc around the binding neighbour, which removes only the outward component and keeps the tangential one (a drone can still circle at constant range). Both controllers use the same projection in each comparison, and connectivity is still never violated. These measurements predate the heuristic tuning of Section 6.5 and use

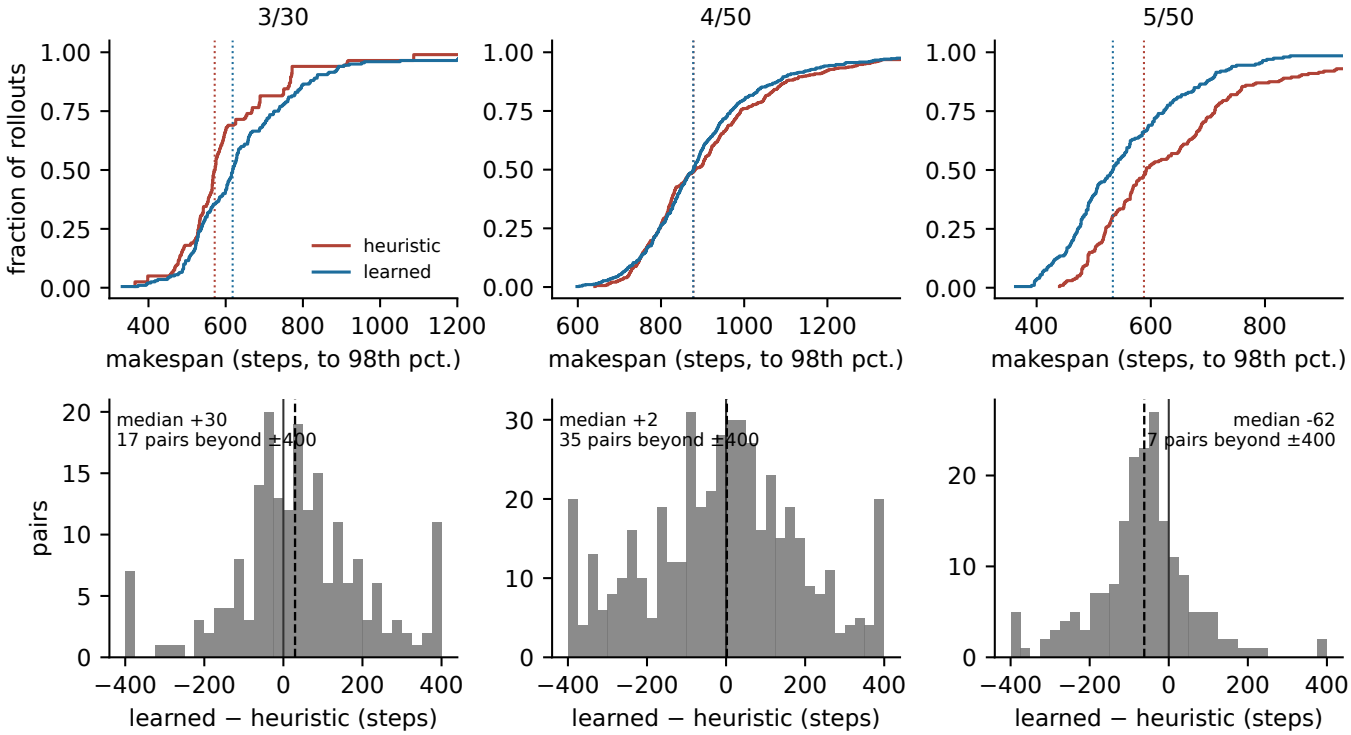


Figure 6: Top: cumulative distribution of makespan on 3/30, 4/50 and 5/50 (dotted lines mark medians; axes cut at the 98th percentile). Bottom: distribution of the paired difference on both-completed instances, clipped to ±400 steps with the number of clipped pairs noted.

$k_{\text{stall}} = 60$, $h_{\text{hold}} = 40$ for both projections, so they are internally consistent. Evaluating the same learned weights under the sliding projection without retraining reduces the 3/30 deficit from +40 to +21; retraining four seeds under it gives +15 to +48 (mean +28) and lowers the learned fleet's stall ratio from 0.27 to 0.17–0.22, with completion still 100%. The heuristic gains little with three drones (median 570 to 571) but gains more than the learned policy with four (866 to 812 against 879 to 850), so the standard-configuration equality becomes +26 in its favour. We therefore keep the radial projection as the primary model and report this as a sensitivity result: about half of the three-drone deficit is a property of how the constraint is enforced rather than of the learner.

Communication radius. Both controllers were run without retraining at $R = 250$ m and $R = 350$ m (Table 7, Figure 7); the learned policy normalises distances by R, and the instance generator rescales the reachable region $0.9NR$ with R, so both controllers see the same instances at each radius. Completion transfers: the learned policy completes every instance at both radii, while the heuristic loses 18 of 200 instances on 5/50 at the shorter radius. Speed shows two effects. The first is the regime shift seen with fleet size: at $R = 250$ m a demand point needs more hops, the fleet is effectively scarcer, and every configuration moves towards the heuristic (+77 on 3/30, +68 on 4/50, and the 5/50 advantage shrinks to an insignificant −16 ($p = 0.33$)); at $R = 350$ m the fleet is effectively more plentiful and the learned policy is faster already with four drones (−40 on 4/50). The second is a training-radius effect: on 3/30 and 5/50 the learned policy's margin is best at the radius it was trained on and worse on both sides (+30 at 300 m against +77 and +52; −62 against −16 and −42), so the normalisation by R transfers completion but not the full speed. The regime variable is therefore the chain demand, the hops a demand point needs relative to the drones available, rather than the fleet size alone, and a deployment at a radius far from the training radius should retrain or at least re-evaluate the boundary.

6.4. RQ4: Upward transfer of the learned policy

The learned policy never saw more than four drones or more than 80 demand points in training, yet it completes every instance with 5 to 12 drones and 50 to 400 demand points and is faster than the heuristic on all of them. Two properties carry this. The set-based encoder has no parameter that depends on N or M, so larger fleets are simply larger sets; and training samples the fleet size and depot afresh every episode. Table 8 separates the two by running the ablation policy that was trained on the fixed 4/50 configuration alone (no configuration randomisation) on the same large-fleet instances without retraining. That policy also completes every instance from 5 to 10 drones, so completion generalises through the architecture alone; but it is slower than the randomised policy on every configuration (no significant difference from the heuristic at 5/50 against −62, and −51 to −174 against −122 to −178 beyond), so speed on large fleets is learned from the variety of small ones. The same

Table 6: Interventions that did not remove the relay-limited deficit, numbered as in the text. Cells give the range over seeds of the paired median difference against the heuristic (positive: learned slower) and, in parentheses, the number of seeds whose difference is significant at $p < 0.05$ (Wilcoxon signed-rank) over the number of seeds trained. The reference row is the proposed policy against the tuned heuristic (Table 5); the variant rows were evaluated against the heuristic with its pre-tuning recovery parameters ($k_{\text{stall}} = 60$, $h_{\text{hold}} = 40$), against which the proposed policy measured +40, −54 and +0, so differences of a few tens of steps between the reference row and a variant row are not meaningful.

#	Intervention (what changed)	3/30	5/50	4/50	Mechanism of failure
—	Proposed configuration (reference)	+30 (1/1)	−62 (1/1)	+0 (0/1)	—
V1	Relay-weighted rule regularisation (×3)	+18, +58 (1/2)	−29, +13 (0/2)	+13, +41 (1/2)	relay-specific regularisation harms
V2	Per-drone relay credit (2.0 / 6.0)	+36 to +82 (4/4)	+60 to +106 (4/4)	+39 to +131 (4/4)	relaying becomes the goal
V3	Checkpoint selection half on 3/30	+37 to +105 (8/8)	−58 to −22 (7/8)	−37 to +108 (3/8)	improves only the standard config.
V4	Zero-sum credit redistribution	+91, +126 (2/2)	+98, +172 (2/2)	+146, +180 (2/2)	same, larger
V5	Joint-action critic	+71 to +124 (4/4)	−46 to −16 (3/4)	+8 to +100 (2/4)	target variance from sampled teammates
V6	Nearest-first decision order	+46, +100 (2/2)	−43, −38 (2/2)	+4, +106 (1/2)	order is not the cause
V7	Re-decision after 60 stalled steps	+82, +95 (2/2)	−38, −34 (2/2)	+17, +55 (0/2)	irregular option lengths
V8	Monotone mixing critic (floor 0.5)	+54 to +230 (4/4)	−28 to +38 (2/4)	+55 to +258 (4/4)	divergence vs. credit on one axis
V9	Sequential single-agent credit	+81 to +213 (4/4)	+8 to +27 (2/4)	+62 to +170 (4/4)	no contraction within a decision
V10	Behaviour-cloning initialisation (± warm-up)	+6 to +69 (3/4)	−42 to −20 (3/4)	−16 to +40 (0/4)	erased within 400 episodes
V11	Bottleneck-arrival penalty (0.15 / 0.25)	+85 to +144 (4/4)	−32 to +13 (2/4)	+7 to +156 (3/4)	reduces the mean, not the maximum

Table 7: Transfer to other communication radii without retraining. Cells give completions (heuristic/learned) and the paired median difference with win–loss and p; $R = 300$ m is the training radius.

N/M	$R = 250$ m	$R = 300$ m	$R = 350$ m
3/30	200/200 / +77 (45–154, $< 10^{-4}$)	200/200 / +30 (81–119, 3.4e-04)	200/200 / +52 (58–141, $< 10^{-4}$)
4/50	118/120 / +68 (40–77, $< 10^{-4}$)	473/480 / +2 (234–239, 0.521)	119/120 / −40 (77–42, 1.1e-04)
5/50	182/200 / −16 (99–83, 0.327)	191/200 / −62 (147–44, $< 10^{-4}$)	200/200 / −42 (152–46, $< 10^{-4}$)

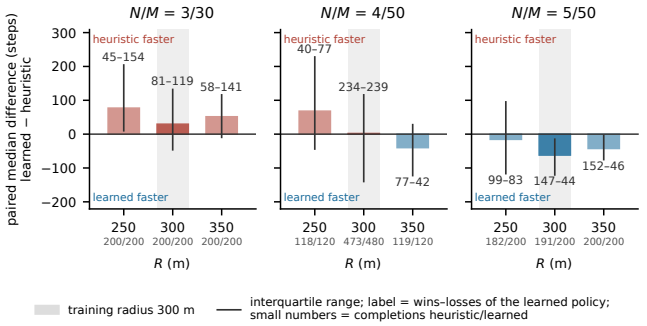


Figure 7: Transfer to other communication radii without retraining, one panel per configuration. Bars are the paired median difference (learned − heuristic, steps): a bar above zero (red) means the heuristic was faster, below zero (blue) the learned policy; whiskers are the interquartile range of the paired differences, the label at each bar is the learned policy's wins–losses, and the completions (heuristic/learned) are printed under the radius. The shaded column is the training radius. Reading left to right within a panel: a shorter radius moves every configuration towards the heuristic, and on 3/30 and 5/50 the learned policy's margin is best at the radius it was trained on.

ablation loses 114–296 steps on 3/30 (Table 10), so randomisation is what makes the small-fleet side usable at all. Both policies were trained with at most four drones; the hop-depth and slot features are normalised by a constant of eight, so fleets of ten and twelve drones also take the encoder outside its training range of feature values, and the policy still completes them.

Where the transfer weakens. We have not isolated the cause of the narrowing at 12/400 and offer three conjectures. First, both controllers maintain a single chain to one lead, so with twelve drones most of the fleet is delivering near the depot regardless of the controller and the decisions that differ between them concern a shrinking share of the work. Second, with 400 demand points a drone's five candidates are all close to it, so the choice among them matters less and the learned policy's advantage, which lies in choosing targets and relays jointly, has less to act on. Third, the manager's hop-depth and slot features are normalised by a constant of eight and exceed one for larger fleets, and twelve sequential autoregressive decisions compound sampling noise; both take the policy farther from its training distribution. Which of these dominates would require fleets beyond twelve drones to settle.

6.5. RQ5: Are the components necessary?

Heuristic. Table 9 grids $k_{\text{stall}} \in \{30, 60, 90, 120\}$, $h_{\text{hold}} \in \{20, 40, 80\}$ and the two recovery modes, plus no recovery, on 3/30, 4/50 and 5/50 with 40 tuning seeds each, and ranks settings by total completions first and the sum of median makespans second; this criterion was fixed before the measurement. The recovery mechanism is necessary: without it the rule completes 25–26 of 40 standard instances (two measurements). The mode matters more than the timing: every lead-swap setting completes 117–120 of 120 instances, while retreat settings complete 101–117 because slackening the chain does not change the assignment that caused the stall. Within the lead-swap family the differences are not statistically meaningful: in a second measurement of the grid, reported in the table together with a paired Wilcoxon test of every setting against the selected one on the same 40 instances, no setting differs from the selected one at $p < 0.05$ on any configuration (all $p \geq 0.09$), and one other setting (30/40) also completes all 120 with a median sum 13 steps lower, which is tie-breaking noise. The selected setting ($k_{\text{stall}} = 120$, $h_{\text{hold}} = 20$) was fixed from the first measurement and is used throughout. The adaptive relay count is necessary as well: fixing one or two relays completes 0–65% (Table 2).

Table 8: Transfer to fleets larger than any seen in training, without retraining. Cells give the learned policy's completions, the paired median difference against the heuristic (negative: learned faster), and in parentheses the learned policy's wins–losses and the Wilcoxon signed-rank p. The proposed policy is trained on $N \in \{3, 4\}$ with random depots; the comparison policy is the ablation trained on the fixed 4/50 configuration only (two seeds).

N/M	trained on $N \in \{3,4\}$, $M \in [20, 80]$	trained on 4/50 only, seed 1	seed 2
5/50	200/200 / −62 (147–44, $< 10^{-4}$)	80/80 / +7 (35–39, 0.840)	80/80 / +44 (28–50, 0.002)
6/80	80/80 / −122 (72–7, $< 10^{-4}$)	80/80 / −56 (57–20, $< 10^{-4}$)	80/80 / −51 (51–27, 0.010)
8/150	80/80 / −158 (75–5, $< 10^{-4}$)	80/80 / −129 (68–11, $< 10^{-4}$)	80/80 / −110 (60–20, $< 10^{-4}$)
10/300	80/80 / −178 (69–11, $< 10^{-4}$)	80/80 / −152 (69–11, $< 10^{-4}$)	80/80 / −174 (68–12, $< 10^{-4}$)

Table 9: Heuristic ablation on tuning seeds disjoint from the evaluation seeds (40 instances per configuration). Cells give completed instances, the median makespan over completed instances and, in parentheses, the p-value of a Wilcoxon signed-rank test of that setting against the selected setting (bold) on the instances both completed; the last column sums completions over the three configurations.

mode	k_{stall} / h_{hold}	3/30	4/50	5/50	completions
none	—	39/40 / 559 (0.228)	25/40 / 817 (0.258)	38/40 / 602 (0.093)	102/120
lead swap	30 / 20	40/40 / 557 (0.121)	39/40 / 853 (0.130)	40/40 / 625 (0.928)	119/120
lead swap	30 / 40	40/40 / 562 (0.673)	40/40 / 867 (0.775)	40/40 / 606 (0.247)	120/120
lead swap	30 / 80	40/40 / 567 (0.158)	39/40 / 844 (0.754)	39/40 / 614 (0.404)	118/120
lead swap	60 / 20	40/40 / 557 (0.599)	39/40 / 849 (0.780)	40/40 / 600 (0.791)	119/120
lead swap	60 / 40	40/40 / 554 (0.497)	39/40 / 852 (0.807)	40/40 / 625 (0.944)	119/120
lead swap	60 / 80	40/40 / 556 (0.359)	39/40 / 863 (0.788)	39/40 / 632 (0.675)	118/120
lead swap	90 / 20	40/40 / 566 (0.206)	39/40 / 871 (0.908)	40/40 / 599 (0.375)	119/120
lead swap	90 / 40	40/40 / 559 (0.473)	40/40 / 856 (0.674)	37/40 / 634 (0.844)	117/120
lead swap	90 / 80	40/40 / 563 (0.786)	39/40 / 858 (0.149)	39/40 / 629 (0.615)	118/120
lead swap	**120 / 20**	40/40 / 561	40/40 / 867	40/40 / 620	**120/120**
lead swap	120 / 40	40/40 / 557 (0.757)	39/40 / 878 (0.685)	40/40 / 624 (0.979)	119/120
lead swap	120 / 80	40/40 / 562 (0.687)	38/40 / 829 (0.546)	39/40 / 626 (0.850)	117/120
retreat	30 / 20	37/40 / 559 (0.350)	35/40 / 836 (0.185)	40/40 / 632 (0.528)	112/120
retreat	30 / 40	39/40 / 556 (0.079)	29/40 / 832 (0.284)	39/40 / 621 (0.157)	107/120
retreat	30 / 80	38/40 / 561 (0.754)	26/40 / 804 (0.443)	37/40 / 584 (0.368)	101/120
retreat	60 / 20	39/40 / 556 (0.866)	38/40 / 847 (0.215)	38/40 / 603 (0.759)	115/120
retreat	60 / 40	39/40 / 559 (0.875)	39/40 / 823 (0.485)	39/40 / 624 (0.459)	117/120
retreat	60 / 80	39/40 / 563 (0.991)	26/40 / 795 (0.236)	40/40 / 635 (0.490)	105/120
retreat	90 / 20	39/40 / 566 (0.512)	38/40 / 822 (0.058)	40/40 / 642 (0.773)	117/120
retreat	90 / 40	39/40 / 566 (0.180)	39/40 / 852 (0.634)	39/40 / 618 (0.615)	117/120
retreat	90 / 80	39/40 / 562 (0.665)	36/40 / 830 (0.578)	37/40 / 598 (0.354)	112/120
retreat	120 / 20	39/40 / 556 (0.645)	37/40 / 863 (0.327)	38/40 / 614 (0.460)	114/120
retreat	120 / 40	39/40 / 560 (0.757)	37/40 / 831 (0.451)	38/40 / 628 (0.602)	114/120
retreat	120 / 80	39/40 / 561 (0.191)	38/40 / 854 (0.815)	37/40 / 627 (0.308)	114/120

Learned policy. Table 10 removes one element at a time, retrains two seeds with the same budget, and runs each ablated policy on the same instances as the proposed policy (the first 120, 80 and 80 paired runs of Section 6.3); cells give the median of the per-instance makespan difference and a Wilcoxon signed-rank p, per seed. Rule regularisation and entropy-target annealing are necessary everywhere: removing either costs 52–189 steps on every configuration and seed at $p < 10^{-4}$. Configuration randomisation is necessary off the training distribution: removing it is neutral on the standard configuration (−19 and −34, $p \geq 0.12$), on which the ablated policy was trained, but costs 114–296 steps on 3/30 and 53–76 on 5/50. Autoregressive decisions help on every configuration by 32–88 steps ($p \leq 0.007$). The residual penalty is neutral: no seed on any configuration differs from the proposed policy at $p < 0.05$. A manager with a fixed-size MLP encoder instead of the set encoder, trained in an earlier version on the standard configuration only, completed 38% of standard instances and could not be run on other fleet sizes at all.

6.6. Inference cost

Table 11 gives the latency of one manager decision on a CPU. The heuristic costs 0.05–0.12 ms; the learned policy 1.3–2.5 ms without autoregression and 2.7–7.4 ms with it, growing linearly with the fleet because each drone is a sequential network call. Over a 900-step episode the learned policy's cumulative decision time is 0.16 s. The exact search takes seconds to 52 minutes on instances with three to five demand points, and the MILP for larger instances uses its full hour.

7. Discussion

Deployment. The two controllers are complementary and the boundary is sharp, so a fleet operator can choose by fleet size:

Table 10: Ablation of the learned policy. Each cell gives, for two independently trained seeds, the median per-instance makespan difference between the ablated policy and the proposed policy on the same instances (positive: the removed element was needed) and the Wilcoxon signed-rank p.

Removed component	4/50 (120 runs)	3/30 (80)	5/50 (80)
Rule regularisation	+121 ($< 10^{-4}$) / +189 ($< 10^{-4}$)	+171 ($< 10^{-4}$) / +160 ($< 10^{-4}$)	+52 ($< 10^{-4}$) / +152 ($< 10^{-4}$)
Configuration randomisation	−19 (0.260) / −34 (0.119)	+114 ($< 10^{-4}$) / +296 ($< 10^{-4}$)	+53 (2.7e-04) / +76 ($< 10^{-4}$)
Entropy target annealing	+136 ($< 10^{-4}$) / +138 ($< 10^{-4}$)	+138 ($< 10^{-4}$) / +130 ($< 10^{-4}$)	+58 ($< 10^{-4}$) / +76 ($< 10^{-4}$)
Autoregressive decisions	+32 (0.007) / +84 ($< 10^{-4}$)	+88 (3.2e-04) / +56 (0.002)	+42 (2.7e-04) / +60 (9.1e-04)
Residual penalty	−4 (0.746) / −32 (0.204)	+16 (0.625) / −28 (0.291)	+24 (0.120) / +12 (0.374)

Table 11: Latency of one manager decision (milliseconds, CPU) and the cumulative decision time of a 900-step episode.

Controller	$N = 3$	$N = 4$	$N = 6$	$N = 8$	900 steps
Geometric rule + recovery	0.05	0.08	0.10	0.12	0.003 s
Learned, independent decisions	1.27	1.52	1.94	2.45	0.069 s
Learned, autoregressive (proposed)	2.70	3.52	5.32	7.35	0.158 s

the heuristic with three drones, the learned policy with five or more, where it is both faster and more reliable; with four the two are statistically indistinguishable in speed and the learned policy completes more. Both decide in milliseconds and both keep the fleet connected by construction.

What the negative results mean. Eleven interventions that change how the learner is credited, informed, initialised or rewarded leave the relay-limited deficit in place, and a change to the environment's enforcement model removes half of it. The deficit is therefore best understood as an interaction between the learner's fixed point under this objective and the geometry of the constraint, not as a shortcoming that more training or a better critic would fix. For a practitioner the actionable consequence is the one above: pick the controller by regime.

Limitations. The learned policy was trained at a single communication radius; it transfers to radii of 250 and 350 m in completion but its speed ranking against the heuristic shifts with the radius (Section 6.3), and radii outside this range were not tested. The environment is planar, ignores wind, battery and collisions, and uses a disc communication model; real links are not discs. The exact optimum is that of a discretised model: its lattice spacing and ten-step periods leave a quantisation that the finest lattice we could afford does not remove, and for instances that need reloading only solver incumbents with a remaining optimality gap are available. The heuristic's novelty lies in the recovery mechanism and the adaptive relay count rather than in the chain geometry itself.

Methodological notes. Three practices mattered for the conclusions. Completed-episode averages favour the controller that gives up, so comparisons must be paired. Forty runs cannot support claims about tails; several of our own interim conclusions reversed when the sample grew. And a pre-check that uses a different metric from the final judgement will pass interventions that the final judgement rejects.

8. Conclusion

We proposed two controllers for connectivity-constrained drone delivery, a geometric chain-relay heuristic with deadlock recovery and a fleet-size-agnostic learned policy, both acting through a projection that keeps the fleet connected at every step. Under a paired protocol across fleets of three to twelve drones the learned policy completes every instance and is faster by 38 to 178 steps once more than four drones are available, although it was trained on three and four, while the heuristic is faster by 30–39 steps with three. The relay-limited deficit is one decision, which drone relays; it resists eleven policy-side interventions and halves when the enforcement model is changed. Where an exact optimum of the space–time model is available, the learned policy is within 2–17% of it and the heuristic within 11–23%, and both decide three to six orders of magnitude faster than the exact methods, which stop proving optimality once reloading is needed. The natural next step is a policy whose output is a team assignment rather than a set of per-drone options, so that the relay identity becomes a coordinate the critic can value directly.

Acknowledgements

[To be added.]

Data and code availability

Code, trained weights, instance generators and the paired-comparison records are available at `https://github.com/koptimizer/relay_drone_network`.

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